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C01 - ASSESSMENT TOOL – 1– APPLICATION BASED QUESTIONS

COURSE CODE:DSA0308

COURSE NAME:- Natural Language Processing

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Question 1:- Resume Information Extraction System Using Regular Expressions

Explanation:

1. First, we import the re library to use regular expressions in Python.

2. We ask the user to enter the candidate's resume details.

3. The program searches for the candidate's name in the given text.

4. It finds the email address from the resume.

5. It extracts the candidate's mobile number using a regular expression.

6. The program checks which technical skills are mentioned in the resume, such as Python, Java, SQL, Machine Learning and NLP

7. We use re.search() to find the required information in the text.

8. The program also finds the number of years of experience mentioned in the resume.

9. All the extracted details are displayed on the screen.

10. Next, the program checks whether the candidate has Python skills.

11. It also checks whether the candidate has at least two years of experience.

12. If both conditions are satisfied, the candidate is marked as eligible.

13. If either condition is not satisfied, the candidate is marked as not eligible.

14. Finally, the program displays the candidate's profile with the eligibility result.

Input

Enter the following resume text when the program asks Enter resume text:.

Ananya Sharma is a software developer with 3 years of experience. Her email is ananya@gmail.com and her mobile number is 9876543210. She has skills in Python, SQL, NLP, Java and Machine Learning

Output:-

Name: Ananya Sharma

Email: ananya@gmail.com

Mobile: 9876543210

Skills: ['Python', 'Java', 'SQL', 'Machine Learning', 'NLP']

Experience: 3 years

Eligible: Yes

3. Result:- The resume information extraction system was implemented successfully using Python Regular Expressions. It extracts the candidate's name, email address, mobile number, technical skills, and years of experience. The candidate is marked as eligible because they have Python skills and three years of experience, which satisfies the minimum eligibility criteria.

Question 2:- Regular Expressions – Product Search System

Explanation:

1. First, we import the re library to perform product searches using Regular Expressions.

2. We create a list of product names available in the e-commerce store.

3. The user enters a keyword to search for products.

4. Exact search finds a product whose complete name matches the entered keyword.

5. Prefix search finds products whose names start with the given keyword, while suffix search finds products whose names end with it.

6. Partial search finds products that contain the keyword anywhere in their names.

7. Case-insensitive search matches products without considering uppercase or lowercase letters.

8. We use re.search() to check product names against the search patterns, and re.escape() to handle special characters safely.

9. The program stores all matching product names in a list and uses the len() function to count the matches for each search.

10. Finally, the program displays the matching product names and the total number of matching products for all five search methods.

Input:-

Enter the following keyword when the program asks Enter product keyword:

Apple

Output:-

Enter product keyword: Apple

Exact Search

Matching Products: []

Total Matches: 0

Prefix Search

Matching Products: ['Apple iPhone', 'Apple Laptop']

Total Matches: 2

Suffix Search

Matching Products: []

Total Matches: 0

Partial Search

Matching Products: ['Apple iPhone', 'Apple Laptop']

Total Matches: 2

Case-Insensitive Search

Matching Products: ['Apple iPhone', 'Apple Laptop']

Total Matches: 2

Result:-The product search system was implemented successfully using Python Regular Expressions. It performs exact, prefix, suffix, partial keyword, and case-insensitive searches. The program displays the matching product names and the total number of matching products for each search method.

Question 3: Regular Expressions – University Registration System

Explanation:

1. First, we import the re library to validate student information using Regular Expressions.

2. The program asks the student to enter their register number, institutional email, course code, semester, and mobile number.

3. The register number is checked to ensure it follows the required university format.

4. The institutional email address is checked to ensure it follows the correct email format and uses the university's domain.

5. The course code is validated to check whether it follows the required format.

6. The semester information is checked to ensure that the student enters a valid semester number.

7. The mobile number is checked to ensure that it contains a valid 10-digit Indian mobile number.

8. Regular Expressions are used to compare each entered value with its required pattern.

9. The program displays a separate validation message for each field, indicating whether the information is valid or invalid.

10. Finally, the program displays the registration status as successful if all fields are valid; otherwise, it displays registration failed.

Sample input:-

Enter register number: 1923240871

Enter institutional email:simats@university.edu

Enter course code:DSA0308

Enter semester (1-8): 6

Enter mobile number: 9876543210

Output:-

Register Number: Valid

Institutional Email: Valid

Course Code: Valid

Semester: Valid

Mobile Number: Valid

Registration Status: Successful

Result:- The university registration validation system was implemented successfully using Python Regular Expressions. It validates all five student information fields, displays appropriate validation messages, and generates the final registration status based on the validation results.